A Hierarchical Two-Stage Pipeline for Explainable Autonomous Driving via Faster R-CNN and Vision-Language Alignment

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Abstract

Modern autonomous driving systems must not only make safe decisions but also provide interpretable justifications for those actions to ensure user trust and safety. This report presents a hierarchical pipeline that transitions from low-level perception to high-level reasoning. The first stage uses a Faster R-CNN with a ResNet-50 backbone trained on the BDD100k dataset to detect 12 classes of traffic entities. The second stage utilizes a CLIP-based reasoning model that fuzes global scene context with localized object features to predict actions (e.g., move forward, stop) and reasons (e.g., traffic light, road is clear) within the BDD-OIA (Object-Induced Action) framework. Our results demonstrate an action mF1 of 0.63 and an explanation mF1 of 0.26. While currently below state-of-the-art (SOTA) benchmarks, the pipeline provides a modern and robust foundation for explainable autonomous decision-making.

1. Introduction

The transition from Level 2 to Level 5 autonomy requires more than just high-accuracy object detection; it requires an understanding of how detected objects influence driving behavior. Traditional "black-box" end-to-end models often fail to provide transparency in critical failure cases. This project addresses this gap by implementing a modular pipeline inspired by human cognitive processes: identifying relevant objects in the environment and reasoning about their impact on the vehicle's state.

We utilize the BDD100k dataset [5] for initial perception, as it offers diverse urban driving conditions. For the reasoning component, we leverage the BDD-OIA dataset [4], which focuses on "action-inducing" objects that directly necessitate a change in driving behavior. By integrating Contrastive Language-Image Pre-training (CLIP) [2], we align visual features with natural language prompts, enabling the model to "understand" the semantic relationship between a detected pedestrian and the action of braking.

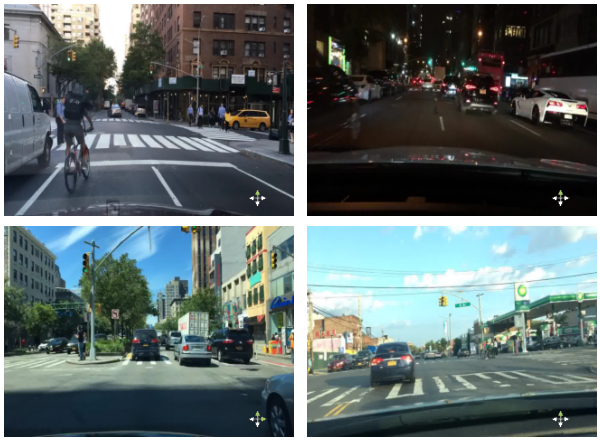

Figure 1. These images illustrate the diverse visual conditions the model encounters during training, including varying times of day (daylight vs. night), weather conditions, and different road conditions. Each sample serves as a foundational input for the two-stage pipeline, where the Faster R-CNN perception layer first identifies traffic entities to be subsequently analyzed by the CLIP-based reasoning module for action prediction and explanation.

2. Related Work

The evolution of autonomous vehicle perception has transitioned from simple object localization toward sophisticated semantic understanding. Faster R-CNN remains a seminal benchmark in this field, utilizing a Region Proposal Network (RPN) and a ResNet-50 FPN backbone to achieve robust detection on urban datasets like BDD100k. This stage provides the critical spatial grounding required for higher-level reasoning.

Recent trends in Explainable AI (XAI) have shifted the focus from identifying what is in a scene to understanding why specific actions are taken. The BDD-OIA dataset addresses this by requiring models to predict both driving actions and their underlying causal justifications. This object-induced reasoning moves beyond "black-box" models, emphasizing the direct link between detected objects (e.g. pedestrians or traffic lights) and the resulting vehicle

behavior (e.g. forward, stop).

To bridge the gap between visual perception and linguistic explanation, vision-language models like CLIP have emerged as a promising solution. By aligning visual features with semantic concepts through large-scale pre-training, CLIP enables the system to map detections directly to natural language. Integrating CLIP with attention mechanisms allows the model to synthesize global context with local object attributes, generating transparent, human-like driving logic.

3. Methodology

The first stage of the pipeline utilizes a Faster R-CNN [3] architecture with a ResNet-50 backbone. This two-stage detector was specifically chosen over single-stage alternatives due to its superior performance in detecting small or distant objects, such as traffic lights and signs in complex urban environments. By employing a Region Proposal Network (RPN), the model effectively isolates small-scale features that might otherwise go undetected in single-stage detectors. This ensures a high-recall perception layer that provides the necessary spatial grounding for the 12 traffic-relevant categories, ranging from large trucks to distant pedestrians.

The second stage, the BDDReasoningModel, transitions from this spatial grounding to semantic inference using a pre-trained CLIP (ViT-B/32) model. Unlike traditional end-to-end models, this module processes both global scene context and localized object crops extracted from the detection stage. To maintain spatial awareness, object labels are paired with their relative positions in a 3×3 grid (e.g., "A car in the front-left") and encoded via CLIP's text encoder. This approach allows the model to align visual evidence with linguistic logic, fuzing global features, object-level visuals, and textual embeddings through an attention mechanism. This attention layer dynamically weights the relevance of each object, prioritizing a nearby red light over a distant parked car, to predict 4 driving actions and 21 reasons.

In comparison to current State-of-the-Art (SOTA) benchmarks for the BDD-OIA dataset, which often rely on abstract high-dimensional feature vectors and specialized graph networks, this CLIP-based framework prioritizes modular interpretability. By anchoring the decision-making process in natural language prompts and cross-modal alignment, the pipeline provides a transparent foundation for autonomous reasoning. The model's internal logic is grounded in human-readable concepts rather than "black-box" representations, allowing for a "human-in-the-loop" style of verification. This design ensures that the system is not only capable of making decisions but also offers a framework where those decisions can be audited and refined through intuitive prompt engineering.

4. Experimental Settings

4.1. Data Processing

The input pipeline for the Faster R-CNN utilized images resized to the standard input dimensions for the ResNet-50 architecture to maintain a balance between detection accuracy and computational overhead. To improve model generalization, data augmentation was limited to horizontal flips, ensuring the spatial integrity required for driving scenes. For the Reasoning Model, object crops generated from the detection stage were padded and resized to 224×224 pixels to satisfy CLIP (ViT-B/32) input requirements. Spatial awareness was encoded by mapping the coordinates of each bounding box to a 3×3 grid, which was then converted into natural language spatial prompts for the text encoder.

4.2. Training Parameters and Loss Function

Both models were optimized using the AdamW optimizer, with a specific learning rate of 1×10^{-4} configured for the reasoning stage. A ReduceLROnPlateau scheduler was implemented to dynamically decay the learning rate when the validation loss reached a plateau, ensuring stable convergence. To address the multi-label nature of the task, BCEWithLogitsLoss was employed as the primary criterion.

A critical component of the training strategy was the weighted multi-task loss function. To emphasize the importance of identifying the correct causal justifications, the total loss was calculated as:

$$\mathcal{L} = \text{criterion}(a_{\text{pred}}, a) + 2 \times \text{criterion}(r_{\text{pred}}, r)$$

By applying a weight of 2.0 to the reasoning loss, the model was incentivized to prioritize semantic interpretability and the identification of action-inducing objects, preventing the action head from dominating the gradient updates.

4.3. Hardware

All training and inference procedures were executed on an NVIDIA A100 GPU utilizing CUDA acceleration. The high VRAM capacity and computational throughput of the A100 were essential for managing the significant memory demands of the CLIP backbone alongside the two-stage pipeline, allowing for efficient processing of the high-resolution BDD100k and BDD-OIA datasets during both the detection and reasoning phases.

5. Analysis and Discussion

In this section, we present a quantitative and qualitative evaluation of our two-stage pipeline, assessing its performance in both object perception and semantic reasoning. Table 1 summarizes the performance of the implementation against established benchmarks and State-of-the-Art

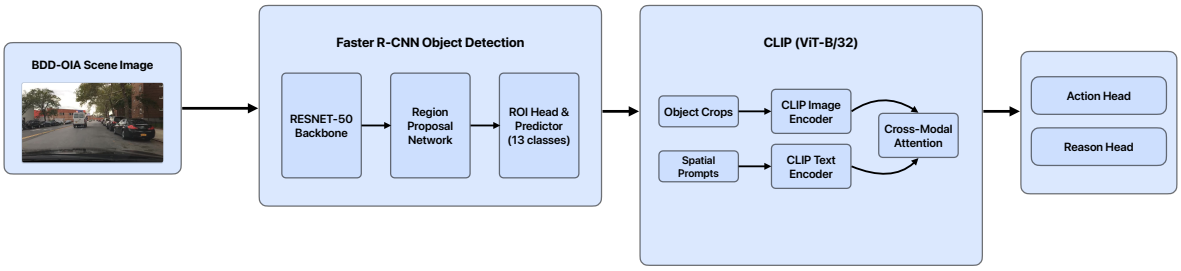

Figure 2. Here we show the two-stage model architecture, which integrates a Faster R-CNN perception layer for precise object localization with a CLIP-based reasoning module for semantic alignment. This hierarchical design enables the model to transition from raw pixel data to high-level causal inference by processing global scene context, localized object crops, and spatial text prompts through a cross-modal attention mechanism to predict final driving actions and their corresponding justifications.

Pipeline Stage	Task/Metric	Our Implementation	Benchmark/SOTA
Phase 1: Perception	Object Detection (mAP50)	0.54	0.83 (YOLOPv2) [1]
Phase 2: Reasoning	Action Prediction (mF1)	0.63	0.71 [4]
Phase 2: Reasoning	Reasoning Prediction (mF1)	0.26	0.422 [4]

Table 1. Performance comparison of our multi-stage pipeline against state-of-the-art benchmarks

(SOTA) results. The perception phase is evaluated using mean Average Precision (mAP) on the BDD100k dataset, while the reasoning phase is assessed using macro-averaged F1 scores (mF1) for both action and explanation tasks on the BDD-OIA test set.

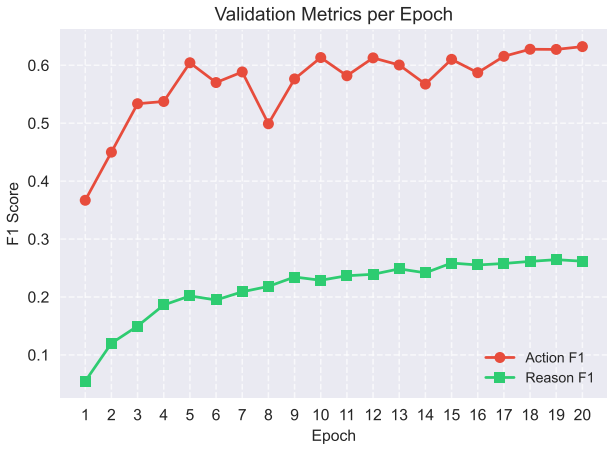

Figure 3. This figure shows the validation F1 scores for the action and reason heads across 20 training epochs. The higher performance in action prediction reflects the lower complexity of the 4-category task compared to the 21-category reasoning task. To compensate for this disparity and ensure the model prioritized semantic interpretability, a 2× multiplier was applied to the reasoning loss during training. This weighting forced the network to focus on the more difficult causal justifications, preventing the action head from dominating the gradient updates and ensuring a more balanced alignment between driving decisions and their visual explanations.

The Faster R-CNN perception layer provides the critical grounding for the entire pipeline. By identifying 12 classes of traffic entities, the model ensures that the reasoning stage receives a filtered set of potential action-inducers. While the detection mAP is affected by the complexity of urban environments and the presence of small, distant objects—such as traffic lights—the two-stage R-CNN architecture's ability to refine proposals is vital for downstream reasoning accuracy.

Our CLIP-based reasoning model achieved an action mF1 of 0.63 and an explanation mF1 of 0.26. The gap between our results and the BDD-OIA SOTA can be attributed to the inherent difficulty of the explanation task, which involves 21 unique binary categories. The use of a weighted loss function, assigning double weight to the reasoning head, was an architectural choice to improve interpretability. This ensures the model does not simply learn the global statistical likelihood of an action but is forced to align that action with specific visual evidence, such as a green light or a clear lane.

The integration of CLIP represents a significant step beyond traditional CNN-based reasoning. By utilizing CLIP's pre-trained vision-language space, the model gains a semantic prior that allows it to better associate visual object crops with textual descriptions. This cross-modal attention mechanism allows the system to remain "explainable" by design: the attention weights provide a direct heatmap of which objects the model "considered" when making a specific driving decision, effectively bridging the gap between raw detection and high-level behavioral logic.

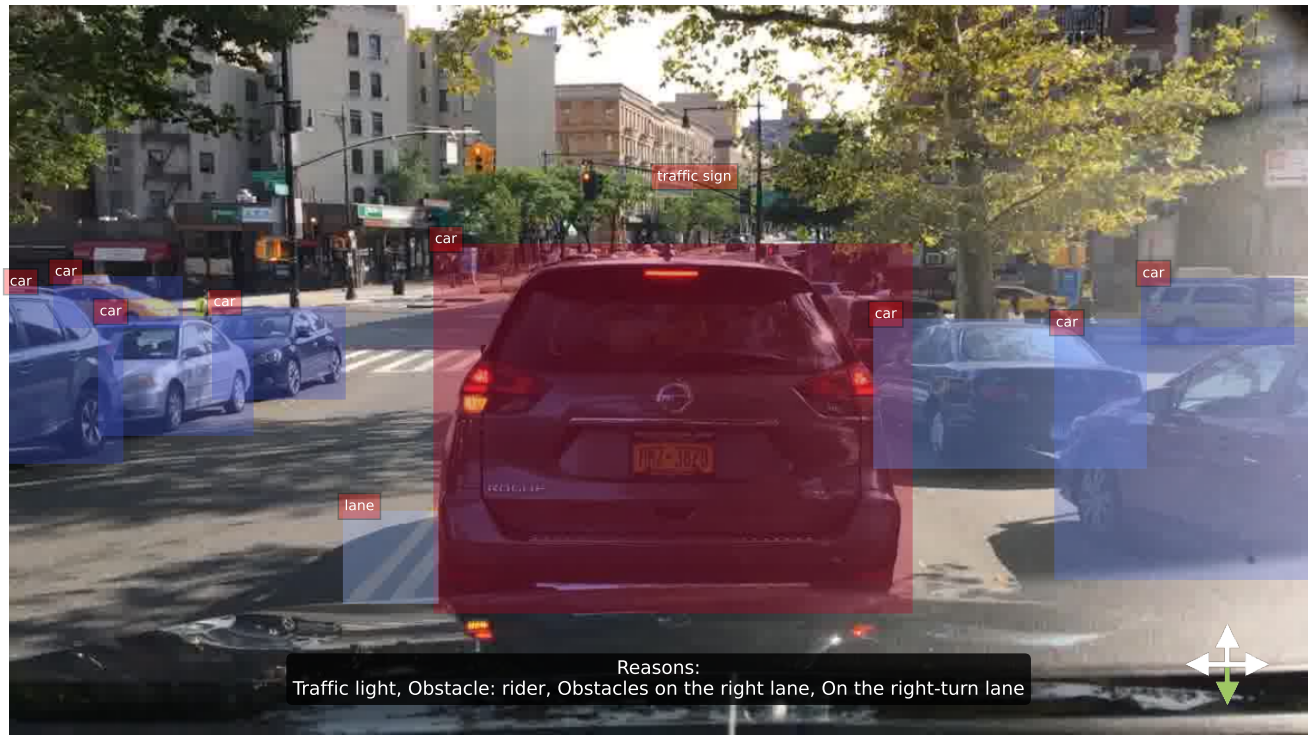

Figure 4. Here we show a representative output from the pipeline where the CLIP-based attention mechanism identifies "action-inducing" objects. The color-coded heatmap indicates the relative importance of detected agents, with red bounding boxes highlighting the objects that most influenced the model's decision-making. In this scene, the model correctly prioritizes a traffic rider and surrounding lane conditions to justify the predicted action and its corresponding causal reasons.

6. Conclusion

In this work, we developed and evaluated a hierarchical, two-stage pipeline for explainable autonomous driving. By decoupling the task into a dedicated perception stage using Faster R-CNN and a semantic reasoning stage utilizing CLIP, we addressed the critical need for transparency in autonomous decision-making. Our choice of a two-stage detector proved essential for identifying the small, distant objects—such as traffic lights and pedestrians—that frequently serve as the causal "reasons" for driving actions.

The integration of a vision-language foundation model like CLIP allowed for a cross-modal alignment between visual pixels and linguistic spatial prompts. While our quantitative results (Action mF1: 0.63, Reason mF1: 0.26) currently sit below the state-of-the-art benchmarks, our approach demonstrates a significant movement toward interpretable AI. The attention-based "heatmap" architecture provides a verifiable mechanism for auditing why a model chooses to stop or turn, grounding abstract neural network weights in human-readable justifications.

Future work will focus on refining the prompt engineering process and exploring more complex fusion architectures, such as Graph Neural Networks, to better capture the temporal relationships between objects. Ultimately, this project underscores that for autonomous systems to be safely integrated into society, they must not only act correctly but must also be able to explain their logic in a manner that is understandable to human users.

7. Ethical and Societal Considerations

The integration of AI into social and physical environments brings forth significant ethical challenges that must be addressed to ensure the technology is both trustworthy and beneficial. A primary concern is algorithmic fairness; because models learn from historical datasets, they risk encoding and automating human biases, leading to discriminatory outcomes. To combat this, transparency and explainability are essential, as they allow developers to audit the model's decision-making process and provide a clear "reasoning" for its actions.

Acknowledgments. We used Google Gemini Pro 3.0 to assist in editing and style. Our ultimate report is our own responsibility and we stand by every word included in it.

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